

Signal Development for Event-Based Surveillance (EBS)

Signal Development

Signal detection is the process of capturing information that may represent events of potential public health importance. A signal is the unverified information considered by an early warning, alert and response (EWAR) system to represent a potential acute health risk, such as an outbreak. Signals may consist of reports of illness or deaths (individual or aggregated), potential exposure to health-related hazards, or the occurrence of natural or man-made disasters. Signals can be detected through any potential information source (health or non-health, informal or official) and by any EBS modality (i.e., community, facility, hotline, media).

When developing a priority list of signals specific for EBS, it is imperative that they are sensitive, rather than specific, to support the detection of unusual or unexpected events not already being captured by existing indicator-based surveillance (IBS) systems. This means, that you should avoid linking EBS signal definitions to specific diseases or conditions that are already included on a nationally notifiable list (e.g. IDSR, etc.), focusing EWAR efforts on the unknown and unusual over the expected. While signal definitions can also be developed for indicator-based surveillance (IBS) systems – these most often relate to alert or outbreak thresholds for specific diseases or conditions being routinely monitored. This is also an important part of EWAR and epidemic intelligence – however, is not the focus of this specific guidance. The below are important considerations for EBS signal list development and examples of signals.

Example Universal Signals

The below are example signals for EBS which can be considered universal or agnostic in that they are not specific to a particular modality or sector. These signals are intended to be as sensitive as possible to

capture all potential events which may be occurring. They are written in simple language so that they are easy to understand and remember by all implementing EBS, including community members. Another benefit of a universally, agnostic signal list is that it allows for comparison of signal detection between EBS modalities.

- Unusual or unexpected cluster of disease or death (human or animal)
- Severe illness or death in human(s) after contact with a sick or dead animal(s)
- Any health-related event that raises concern, fear, and alarm
- Unusual or unexpected change to the physical quality of the air, water, or soil
- Occurrence of an environment hazard

Example Modality-Specific Signals

In some cases, it may be helpful or necessary to develop signals for specific EBS modalities in addition to universal signals. For example, signals related to detecting events within a laboratory setting may not be appropriate for detecting events within a community. When developing modality-specific signals, it is still important to ensure that they are sensitive and simple, so they remain easy to understand and remember. When considering whether modality-specific signals should be added to a universal signal list, be sure that the more specific signal would not be picked up under one of the signals on the universal signal list. Below are some examples of signals which may be useful in specific settings.

Healthcare Facility-specific Signals (human or animal)

- One or more patients with similar illness that do not respond to standard treatment
- One or more healthcare worker(s) with severe illness after attending to patients with similar symptoms
- Large, unexpected, sudden increases in admissions for any illness of the same type, including patients in intensive care units

Laboratory-specific Facility Signals (human or animal):

- Detection of a novel, emerging, or re-emerging pathogen; or a new and/or unreported strain of an already known pathogen (i.e., increase in positivity rate, new genetic variation, novel resistance profile, etc.)
- Detection of a pathogen in an unusual species (e.g., avian influenza in a mammal)
- Large/sudden or unexpected increases in the number of specimens with the same testing request, or are positive for the same pathogen (including pathogens resistant to multiple antibiotics)

Animal Facility-specific Signals (animal clinic, farm, game/reserve/park, zoo, etc.)

- Cluster of animals presenting with unusual signs or behaviors (e.g., aggression, bleeding, dizziness, weight loss, isolation from other animals, diarrhea, body swellings, lameness, loss of hair or limbs, coughing, excessive drooling, blindness, etc.)

- Cluster of animals exhibiting production losses (e.g., milk, eggs, abortions)
- Severe illness in veterinarian or animal care staff after contact (e.g., culling, feeding, treating, vaccinating) with a sick or dead animal(s)

Environment-type Facility Signals

- Improper waste disposal, leakage, or spillage on land, in air or water bodies
- Unusual change in physical water quality parameters of drinking water sources (e.g., color, taste, odor, suspended solids, turbidity)
- Sudden increase in average atmospheric temperatures
- Sudden increase in vector populations from entomological surveillance

POE Facility Signals

- Cluster of people presenting with similar symptoms and a recent (<2 months) history of travel

Other Facility Type Signals

- Unexpected large numbers of children/students absent from school due to the same illness
- Any cluster of unusual or unexpected illness occurring in the same congregate setting (e.g., refugee/migrant camps, prisons, etc.)
- Unexpected large numbers of pharmacy sales among persons buying medicines for the same illness
- Commonly reported lesions during meat inspection from abattoir and/or in slaughterhouse